# Practice workbook · Prepare cheek-cell or blood slides for a light microscope

Obtaining cheek cells or blood, smearing, staining and mounting for viewing; ordering the steps or explaining why cells were invisible.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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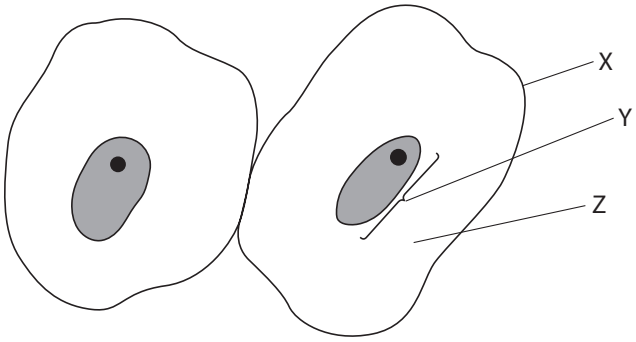
Type ID: HBI-PRAC-PREPARE-MICROSCOPE-SLIDE

# 1. Question 7 · part b · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part b only.

**7** The diagram shows two cheek cells from the inside of the mouth.




(a) Complete the table by giving the missing information.

**(6)**

| Part | Name of part | Function of part |
|---|---|---|
| X | | |
| Y | | |
| Z | | |

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(b) Describe how a student would obtain and prepare these cells so the student can view them using a light microscope.

**(4)**

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# 1. Question 7 · part b · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part b only.

(c) Name two other structures inside the cells that could be seen if an electron microscope is used.

**(2)**

1 .....

2 .....

**(Total for Question 7 = 12 marks)**

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Working and answer

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## 2. Question 3 · part a.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

**3** (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

**(2)**

| | **Steps** |
|---|---|
| | select a suitable objective lens |
| 5 | adjust focusing wheel to obtain a clear image |
| 1 | remove cells from the inside of the cheek |
| | place microscope slide onto the microscope stage |
| | smear the cell sample onto a microscope slide |

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

**(1)**

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

**(1)**

.....

.....

(iv) Give two ways that the student could avoid health hazards when working with living cells.

**(2)**

1 .....

.....

2 .....

.....

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6

P 6 2 0 5 2 A 0 6 2 4

## 2. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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# 3. Question 3 · part a.iii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a.iii only.

**3** (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

**(2)**

| | **Steps** |
|---|---|
| | select a suitable objective lens |
| 5 | adjust focusing wheel to obtain a clear image |
| 1 | remove cells from the inside of the cheek |
| | place microscope slide onto the microscope stage |
| | smear the cell sample onto a microscope slide |

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

**(1)**

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

**(1)**

.....

.....

(iv) Give two ways that the student could avoid health hazards when working with living cells.

**(2)**

1 .....

.....

2 .....

.....

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## 3. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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## 4. Question 6 · part b · 3 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

**6** The lining of the inside of the cheek is made of squamous epithelial cells.

(a) Draw a labelled diagram to show a cheek cell when seen using a light microscope.

**(4)**

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(b) Describe how a sample of cheek cells would be prepared to view using a light microscope.

**(3)**

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## 4. Question 6 · part b · 3 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

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(c) State three differences between a cheek cell and a cell from the lining of the trachea.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

**(Total for Question 6 = 10 marks)**

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Working and answer

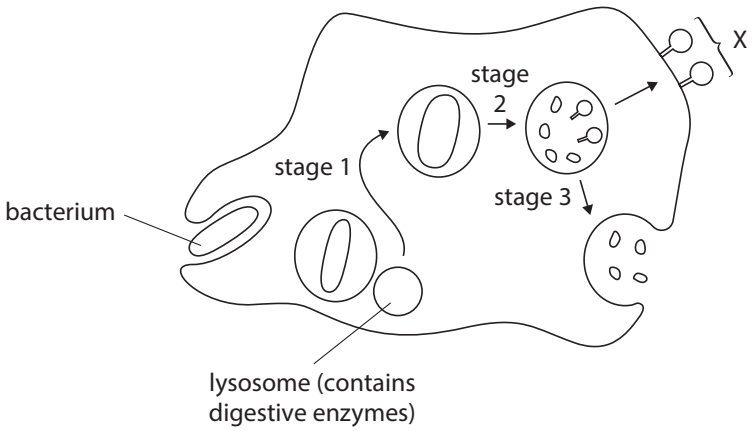
## 5. Question 6 · part b.i · 3 marks

June 2023 Paper 1

Further practice · Attempt part b.i only.

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**6** The diagram shows a white blood cell engulfing a bacterium and the stages that follow.




(a) (i) State the name of the process used by white blood cells to engulf bacteria.

**(1)**

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

**(3)**

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

**(2)**

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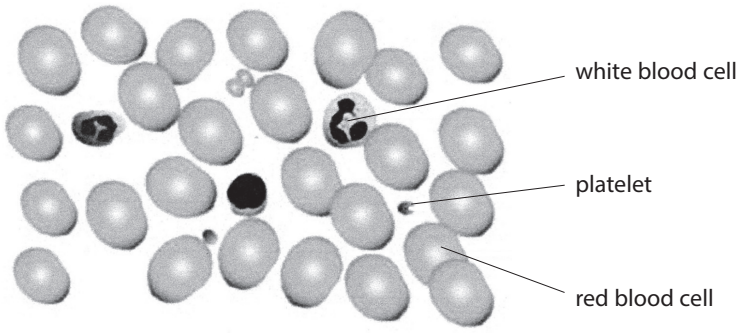
## 5. Question 6 · part b.i · 3 marks

June 2023 Paper 1

Further practice · Attempt part b.i only.

(b) A student observes a prepared slide of blood under a light microscope.

The student draws this diagram of what could be seen.




(i) Describe how the slide of human blood could be prepared.

**(3)**

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## 5. Question 6 · part b.i · 3 marks

June 2023 Paper 1

Further practice · Attempt part b.i only.

(ii) Discuss the accuracy of the student's diagram.

**(4)**

(iii) Explain why the student would be advised to use a prepared slide rather than making one from their own blood.

**(2)**

**(Total for Question 6 = 15 marks)**

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## 5. Question 6 · part b.i · 3 marks · Working and answer

Working and answer

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# 6. Question 7 · part b · 5 marks

June 2025 Paper 2

Further practice · Attempt part b only.

**7** (a) Two children, M and N, are born to two sets of parents.

Child M has blood group O and child N has blood group A.

The blood groups of the two sets of parents are given in the table.

| | Blood group | |
|---|---|---|
| | **mother** | **father** |
| 1st set of parents | AB | O |
| 2nd set of parents | B | B |

(i) Explain which child belongs to which set of parents.

**(4)**

(ii) Calculate the probability that the next child produced by the 2nd set of parents will have blood group B.

Include a genetic diagram in your answer.

**(3)**

probability = .....

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# 6. Question 7 · part b · 5 marks

June 2025 Paper 2

Further practice · Attempt part b only.

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(b) Describe how to prepare a sample of blood to be observed using a light microscope.

**(5)**

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**(Total for Question 7 = 12 marks)**

## 6. Question 7 · part b · 5 marks · Working and answer

Working and answer

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